

Thermodynamics

Instructor

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Chapter 1 – Problems

**Problem numbers correspond to
5th Edition**

1-59



Both a gage and a manometer are attached to a gas tank to measure its pressure. If the reading on the pressure gage is 80 kPa, determine the distance between the two fluid levels of the manometer if the fluid is (a) mercury ($\rho = 13,600 \text{ kg/m}^3$) or (b) water ($\rho = 1000 \text{ kg/m}^3$).

Properties The densities of water and mercury are given to be $\rho_{\text{water}} = 1000 \text{ kg/m}^3$ and $\rho_{\text{Hg}} = 13,600 \text{ kg/m}^3$.

Analysis The gage pressure is related to the vertical distance h between the two fluid levels by

$$P_{\text{gage}} = \rho g h \longrightarrow h = \frac{P_{\text{gage}}}{\rho g}$$

(a) For mercury,

$$\begin{aligned} h &= \frac{P_{\text{gage}}}{\rho_{\text{Hg}} g} \\ &= \frac{80 \text{ kPa}}{(13,600 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \left(\frac{1 \text{ kN/m}^2}{1 \text{ kPa}} \right) \left(\frac{1000 \text{ kg/m} \cdot \text{s}^2}{1 \text{ kN}} \right) = \mathbf{0.60 \text{ m}} \end{aligned}$$

(b) For water,

$$h = \frac{P_{\text{gage}}}{\rho_{\text{H}_2\text{O}} g} = \frac{80 \text{ kPa}}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \left(\frac{1 \text{ kN/m}^2}{1 \text{ kPa}} \right) \left(\frac{1000 \text{ kg/m} \cdot \text{s}^2}{1 \text{ kN}} \right) = \mathbf{8.16 \text{ m}}$$

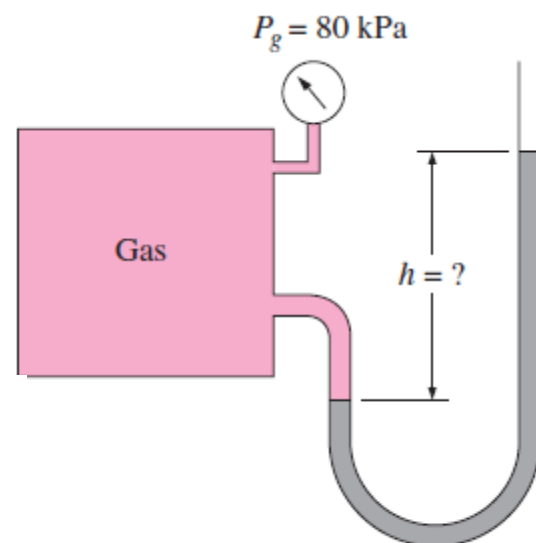


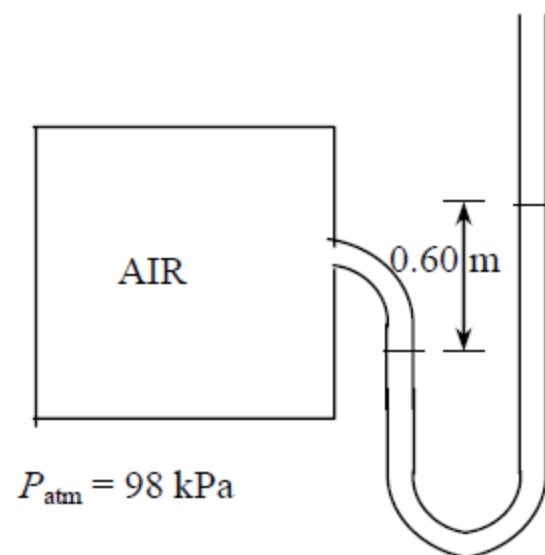
FIGURE P1-59

1-61 A manometer containing oil ($\rho = 850 \text{ kg/m}^3$) is attached to a tank filled with air. If the oil-level difference between the two columns is 60 cm and the atmospheric pressure is 98 kPa, determine the absolute pressure of the air in the tank. *Answer: 103 kPa*

Properties The density of oil is given to be $\rho = 850 \text{ kg/m}^3$.

Analysis The absolute pressure in the tank is determined from

$$\begin{aligned} P &= P_{\text{atm}} + \rho gh \\ &= (98 \text{ kPa}) + (850 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.60 \text{ m}) \left(\frac{1 \text{ kPa}}{1000 \text{ N/m}^2} \right) \\ &= \mathbf{103 \text{ kPa}} \end{aligned}$$



1-62 A mercury manometer ($\rho = 13,600 \text{ kg/m}^3$) is connected to an air duct to measure the pressure inside. The difference in the manometer levels is 15 mm, and the atmospheric pressure is 100 kPa. (a) Judging from Fig. P1-62, determine if the pressure in the duct is above or below the atmospheric pressure. (b) Determine the absolute pressure in the duct.

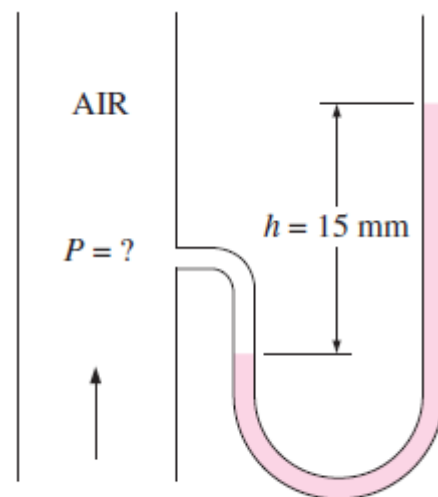


FIGURE P1-62

Properties The density of mercury is given to be $\rho = 13,600 \text{ kg/m}^3$.

Analysis (a) The pressure in the duct is above atmospheric pressure since the fluid column on the duct side is at a lower level.

(b) The absolute pressure in the duct is determined from

$$\begin{aligned}
 P &= P_{\text{atm}} + \rho gh \\
 &= (100 \text{ kPa}) + (13,600 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.015 \text{ m}) \left(\frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) \left(\frac{1 \text{ kPa}}{1000 \text{ N/m}^2} \right) \\
 &= \mathbf{102 \text{ kPa}}
 \end{aligned}$$

1–63 Repeat Prob. 1–62 for a differential mercury height of 45 mm.

Properties The density of mercury is given to be $\rho = 13,600 \text{ kg/m}^3$.

Analysis (a) The pressure in the duct is above atmospheric pressure since the fluid column on the duct side is at a lower level.

(b) The absolute pressure in the duct is determined from

$$\begin{aligned} P &= P_{\text{atm}} + \rho gh \\ &= (100 \text{ kPa}) + (13,600 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.045 \text{ m}) \left(\frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) \left(\frac{1 \text{ kPa}}{1000 \text{ N/m}^2} \right) \\ &= \mathbf{106 \text{ kPa}} \end{aligned}$$

1-66 Consider a 1.8-m-tall man standing vertically in water and completely submerged in a pool. Determine the difference between the pressures acting at the head and at the toes of this man, in kPa.

Assumptions Water is an incompressible substance, and thus the density does not change with depth.

Properties We take the density of water to be $\rho = 1000 \text{ kg/m}^3$.

Analysis The pressures at the head and toes of the person can be expressed as

$$P_{\text{head}} = P_{\text{atm}} + \rho g h_{\text{head}} \quad \text{and} \quad P_{\text{toe}} = P_{\text{atm}} + \rho g h_{\text{toe}}$$

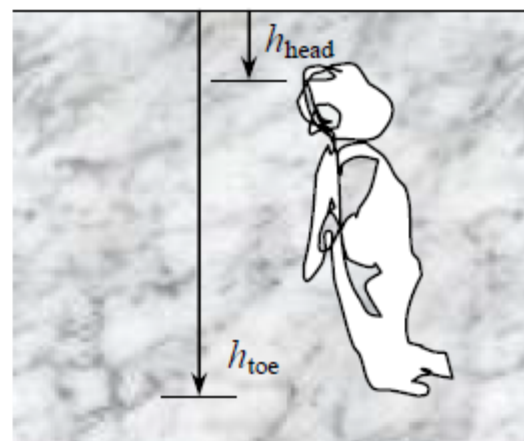
where h is the vertical distance of the location in water from the free surface. The pressure difference between the toes and the head is determined by subtracting the first relation above from the second,

$$P_{\text{toe}} - P_{\text{head}} = \rho g h_{\text{toe}} - \rho g h_{\text{head}} = \rho g (h_{\text{toe}} - h_{\text{head}})$$

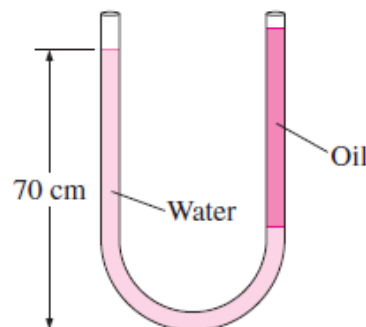
Substituting,

$$P_{\text{toe}} - P_{\text{head}} = (1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(1.80 \text{ m} - 0) \left(\frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) \left(\frac{1 \text{ kPa}}{1000 \text{ N/m}^2} \right) = \mathbf{17.7 \text{ kPa}}$$

Discussion This problem can also be solved by noting that the atmospheric pressure ($1 \text{ atm} = 101.325 \text{ kPa}$) is equivalent to 10.3-m of water height, and finding the pressure that corresponds to a water height of 1.8 m.



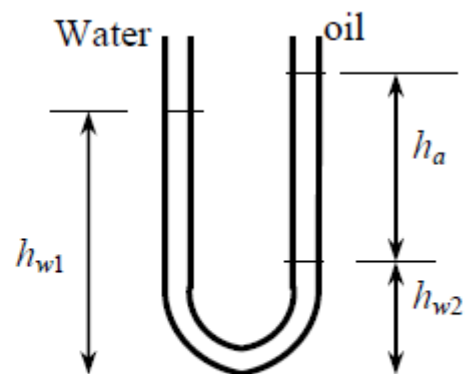
1-67 Consider a U-tube whose arms are open to the atmosphere. Now water is poured into the U-tube from one arm, and light oil ($\rho = 790 \text{ kg/m}^3$) from the other. One arm contains 70-cm-high water, while the other arm contains both fluids with an oil-to-water height ratio of 4. Determine the height of each fluid in that arm.



Assumptions Both water and oil are incompressible substances.

Properties The density of oil is given to be $\rho = 790 \text{ kg/m}^3$. We take the density of water to be $\rho = 1000 \text{ kg/m}^3$.

Analysis The height of water column in the left arm of the manometer is given to be $h_{w1} = 0.70 \text{ m}$. We let the height of water and oil in the right arm to be h_{w2} and h_a , respectively. Then, $h_a = 4h_{w2}$. Noting that both arms are open to the atmosphere, the pressure at the bottom of the U-tube can be expressed as



$$P_{\text{bottom}} = P_{\text{atm}} + \rho_w g h_{w1} \quad \text{and} \quad P_{\text{bottom}} = P_{\text{atm}} + \rho_w g h_{w2} + \rho_a g h_a$$

Setting them equal to each other and simplifying,

$$\rho_w g h_{w1} = \rho_w g h_{w2} + \rho_a g h_a \quad \rightarrow \quad \rho_w h_{w1} = \rho_w h_{w2} + \rho_a h_a \quad \rightarrow \quad h_{w1} = h_{w2} + (\rho_a / \rho_w) h_a$$

Noting that $h_a = 4h_{w2}$, the water and oil column heights in the second arm are determined to be

$$0.7 \text{ m} = h_{w2} + (790/1000) 4h_{w2} \quad \rightarrow \quad h_{w2} = \mathbf{0.168 \text{ m}}$$

$$0.7 \text{ m} = 0.168 \text{ m} + (790/1000) h_a \quad \rightarrow \quad h_a = \mathbf{0.673 \text{ m}}$$

Discussion Note that the fluid height in the arm that contains oil is higher. This is expected since oil is lighter than water.

1–69 Freshwater and seawater flowing in parallel horizontal pipelines are connected to each other by a double U-tube manometer, as shown in Fig. P1–69. Determine the pressure difference between the two pipelines. Take the density of seawater at that location to be $\rho = 1035 \text{ kg/m}^3$. Can the air column be ignored in the analysis?

Assumptions 1 All the liquids are incompressible. 2 The effect of air column on pressure is negligible.

Properties The densities of seawater and mercury are given to be $\rho_{\text{sea}} = 1035 \text{ kg/m}^3$ and $\rho_{\text{Hg}} = 13,600 \text{ kg/m}^3$. We take the density of water to be $\rho_w = 1000 \text{ kg/m}^3$.

Analysis Starting with the pressure in the fresh water pipe (point 1) and moving along the tube by adding (as we go down) or subtracting (as we go up) the ρgh terms until we reach the sea water pipe (point 2), and setting the result equal to P_2 gives

$$P_1 + \rho_w gh_w - \rho_{\text{Hg}} gh_{\text{Hg}} - \rho_{\text{air}} gh_{\text{air}} + \rho_{\text{sea}} gh_{\text{sea}} = P_2$$

Rearranging and neglecting the effect of air column on pressure,

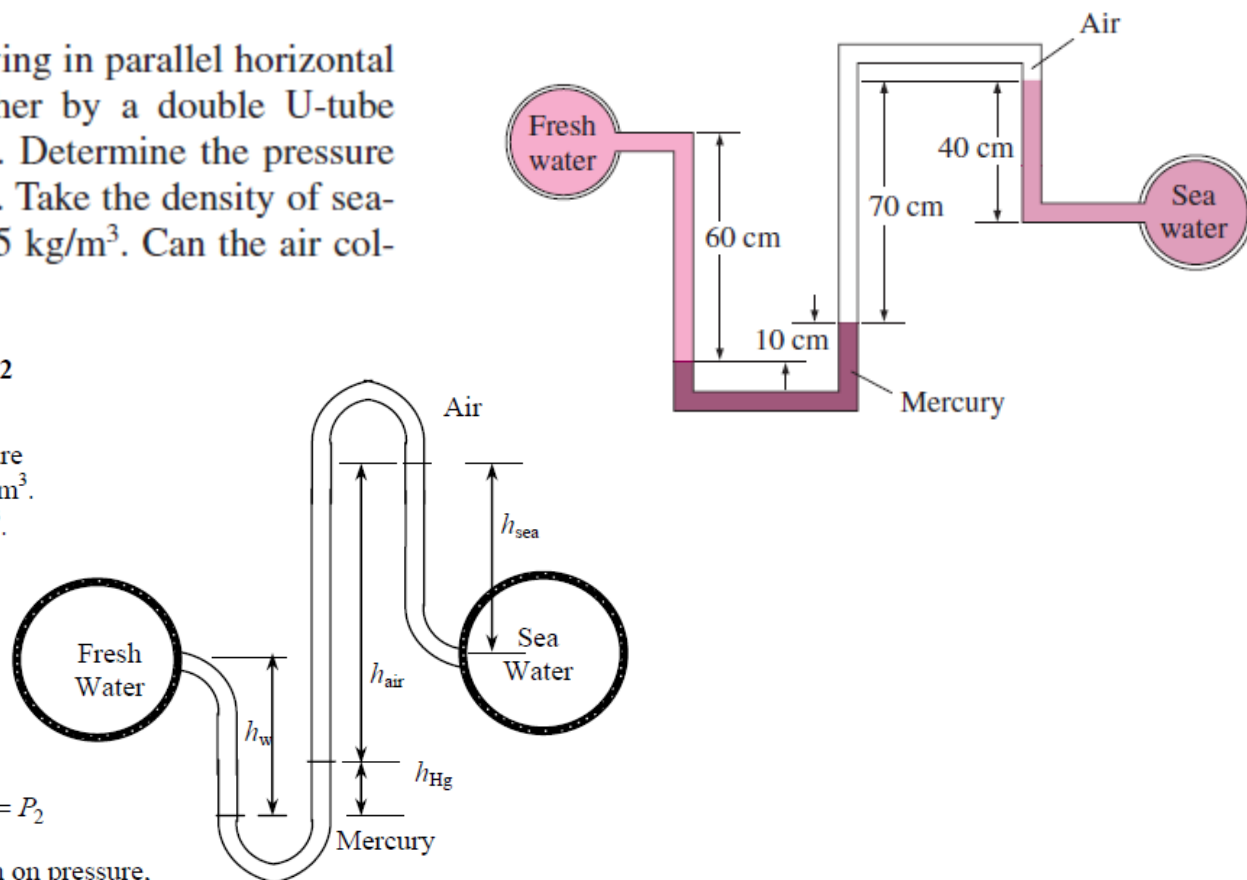
$$P_1 - P_2 = -\rho_w gh_w + \rho_{\text{Hg}} gh_{\text{Hg}} - \rho_{\text{sea}} gh_{\text{sea}} = g(\rho_{\text{Hg}} h_{\text{Hg}} - \rho_w h_w - \rho_{\text{sea}} h_{\text{sea}})$$

Substituting,

$$\begin{aligned} P_1 - P_2 &= (9.81 \text{ m/s}^2)[(13600 \text{ kg/m}^3)(0.1 \text{ m}) \\ &\quad - (1000 \text{ kg/m}^3)(0.6 \text{ m}) - (1035 \text{ kg/m}^3)(0.4 \text{ m})] \left(\frac{1 \text{ kN}}{1000 \text{ kg} \cdot \text{m/s}^2} \right) \\ &= 3.39 \text{ kN/m}^2 = \mathbf{3.39 \text{ kPa}} \end{aligned}$$

Therefore, the pressure in the fresh water pipe is 3.39 kPa higher than the pressure in the sea water pipe.

Discussion A 0.70-m high air column with a density of 1.2 kg/m^3 corresponds to a pressure difference of 0.008 kPa. Therefore, its effect on the pressure difference between the two pipes is negligible.



1-70 Repeat Prob. 1-69 by replacing the air with oil whose specific gravity is 0.72.

Assumptions All the liquids are incompressible.

Properties The densities of seawater and mercury are given to be $\rho_{\text{sea}} = 1035 \text{ kg/m}^3$ and $\rho_{\text{Hg}} = 13,600 \text{ kg/m}^3$. We take the density of water to be $\rho_w = 1000 \text{ kg/m}^3$. The specific gravity of oil is given to be 0.72, and thus its density is 720 kg/m^3 .

Analysis Starting with the pressure in the fresh water pipe (point 1) and moving along the tube by adding (as we go down) or subtracting (as we go up) the ρgh terms until we reach the sea water pipe (point 2), and setting the result equal to P_2 gives

$$P_1 + \rho_w gh_w - \rho_{\text{Hg}} gh_{\text{Hg}} - \rho_{\text{oil}} gh_{\text{oil}} + \rho_{\text{sea}} gh_{\text{sea}} = P_2$$

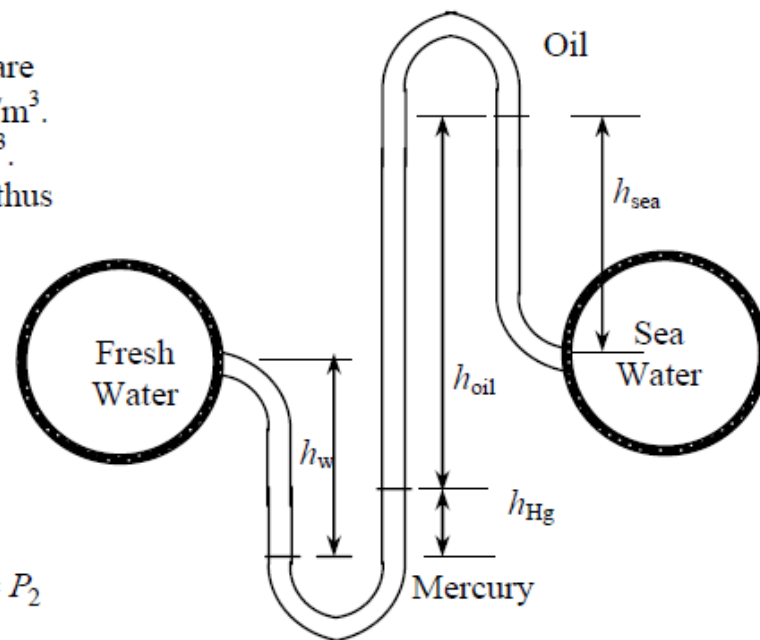
Rearranging,

$$\begin{aligned} P_1 - P_2 &= -\rho_w gh_w + \rho_{\text{Hg}} gh_{\text{Hg}} + \rho_{\text{oil}} gh_{\text{oil}} - \rho_{\text{sea}} gh_{\text{sea}} \\ &= g(\rho_{\text{Hg}} h_{\text{Hg}} + \rho_{\text{oil}} h_{\text{oil}} - \rho_w h_w - \rho_{\text{sea}} h_{\text{sea}}) \end{aligned}$$

Substituting,

$$\begin{aligned} P_1 - P_2 &= (9.81 \text{ m/s}^2)[(13600 \text{ kg/m}^3)(0.1 \text{ m}) + (720 \text{ kg/m}^3)(0.7 \text{ m}) - (1000 \text{ kg/m}^3)(0.6 \text{ m}) \\ &\quad - (1035 \text{ kg/m}^3)(0.4 \text{ m})] \left(\frac{1 \text{ kN}}{1000 \text{ kg} \cdot \text{m/s}^2} \right) \\ &= 8.34 \text{ kN/m}^2 = \mathbf{8.34 \text{ kPa}} \end{aligned}$$

Therefore, the pressure in the fresh water pipe is 8.34 kPa higher than the pressure in the sea water pipe.



1-73 The gage pressure of the air in the tank shown in Fig. P1-73 is measured to be 80 kPa. Determine the differential height h of the mercury column.

Assumptions The air pressure in the tank is uniform (i.e., its variation with elevation is negligible due to its low density), and thus the pressure at the air-water interface is the same as the indicated gage pressure.

Properties We take the density of water to be $\rho_w = 1000 \text{ kg/m}^3$. The specific gravities of oil and mercury are given to be 0.72 and 13.6, respectively.

Analysis Starting with the pressure of air in the tank (point 1), and moving along the tube by adding (as we go down) or subtracting (as we go up) the ρgh terms until we reach the free surface of oil where the oil tube is exposed to the atmosphere, and setting the result equal to P_{atm} gives

$$P_1 + \rho_w gh_w - \rho_{\text{Hg}} gh_{\text{Hg}} - \rho_{\text{oil}} gh_{\text{oil}} = P_{\text{atm}}$$

Rearranging,

$$P_1 - P_{\text{atm}} = \rho_{\text{oil}} gh_{\text{oil}} + \rho_{\text{Hg}} gh_{\text{Hg}} - \rho_w gh_w$$

or,

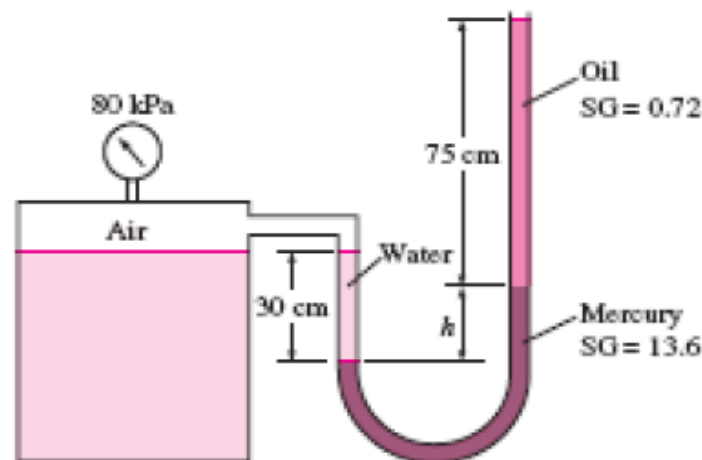
$$\frac{P_{1,\text{gage}}}{\rho_w g} = \text{SG}_{\text{oil}} h_{\text{oil}} + \text{SG}_{\text{Hg}} h_{\text{Hg}} - h_w$$

Substituting,

$$\left(\frac{80 \text{ kPa}}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \right) \left(\frac{1000 \text{ kg} \cdot \text{m/s}^2}{1 \text{ kPa} \cdot \text{m}^2} \right) = 0.72 \times (0.75 \text{ m}) + 13.6 \times h_{\text{Hg}} - 0.3 \text{ m}$$

Solving for h_{Hg} gives $h_{\text{Hg}} = \mathbf{0.582 \text{ m}}$. Therefore, the differential height of the mercury column must be 58.2 cm.

Discussion Double instrumentation like this allows one to verify the measurement of one of the instruments by the measurement of another instrument.



1-74 Repeat Prob. 1-73 for a gage pressure of 40 kPa.

Assumptions The air pressure in the tank is uniform (i.e., its variation with elevation is negligible due to its low density), and thus the pressure at the air-water interface is the same as the indicated gage pressure.

Properties We take the density of water to be $\rho_w = 1000 \text{ kg/m}^3$. The specific gravities of oil and mercury are given to be 0.72 and 13.6, respectively.

Analysis Starting with the pressure of air in the tank (point 1), and moving along the tube by adding (as we go down) or subtracting (as we go up) the ρgh terms until we reach the free surface of oil where the oil tube is exposed to the atmosphere, and setting the result equal to P_{atm} gives

$$P_1 + \rho_w gh_w - \rho_{Hg} gh_{Hg} - \rho_{oil} gh_{oil} = P_{atm}$$

Rearranging,

$$P_1 - P_{atm} = \rho_{oil} gh_{oil} + \rho_{Hg} gh_{Hg} - \rho_w gh_w$$

or,

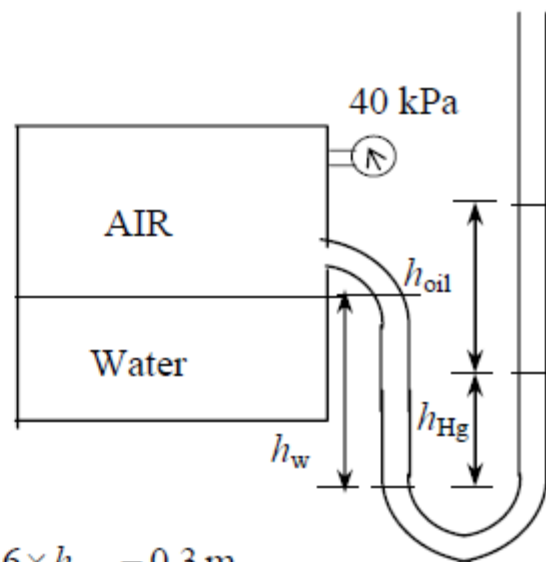
$$\frac{P_{1,gage}}{\rho_w g} = SG_{oil} h_{oil} + SG_{Hg} h_{Hg} - h_w$$

Substituting,

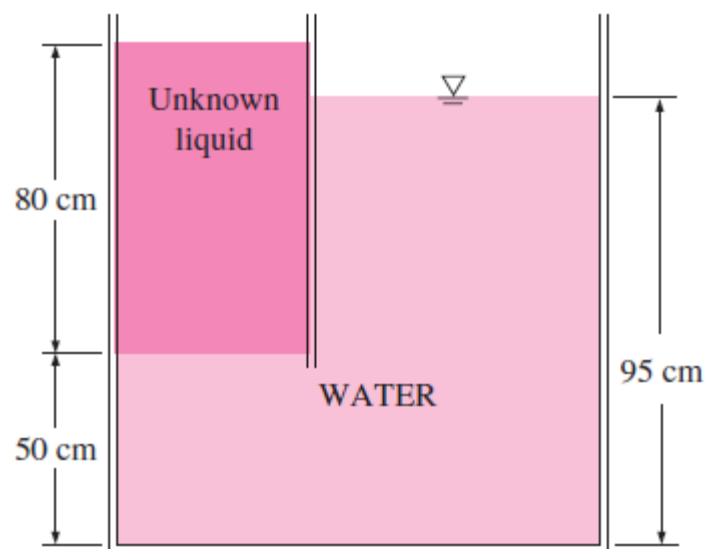
$$\left[\frac{40 \text{ kPa}}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \right] \left(\frac{1000 \text{ kg} \cdot \text{m/s}^2}{1 \text{ kPa} \cdot \text{m}^2} \right) = 0.72 \times (0.75 \text{ m}) + 13.6 \times h_{Hg} - 0.3 \text{ m}$$

Solving for h_{Hg} gives $h_{Hg} = 0.282 \text{ m}$. Therefore, the differential height of the mercury column must be 28.2 cm.

Discussion Double instrumentation like this allows one to verify the measurement of one of the instruments by the measurement of another instrument.



1-75 The top part of a water tank is divided into two compartments, as shown in Fig. P1-75. Now a fluid with an unknown density is poured into one side, and the water level rises a certain amount on the other side to compensate for this effect. Based on the final fluid heights shown on the figure, determine the density of the fluid added. Assume the liquid does not mix with water.



Assumptions 1 Both water and the added liquid are incompressible substances. 2 The added liquid does not mix with water.

Properties We take the density of water to be $\rho = 1000 \text{ kg/m}^3$.

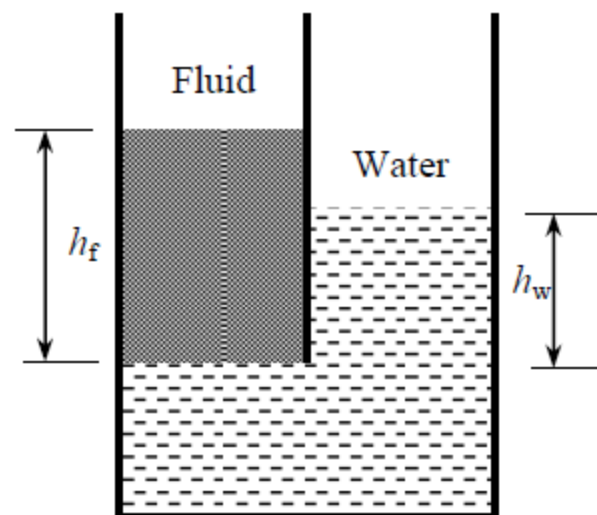
Analysis Both fluids are open to the atmosphere. Noting that the pressure of both water and the added fluid is the same at the contact surface, the pressure at this surface can be expressed as

$$P_{\text{contact}} = P_{\text{atm}} + \rho_f g h_f = P_{\text{atm}} + \rho_w g h_w$$

Simplifying and solving for ρ_f gives

$$\rho_f g h_f = \rho_w g h_w \rightarrow \rho_f = \frac{h_w}{h_f} \rho_w = \frac{45 \text{ cm}}{80 \text{ cm}} (1000 \text{ kg/m}^3) = 562.5 \text{ kg/m}^3$$

Discussion Note that the added fluid is lighter than water as expected (a heavier fluid would sink in water).



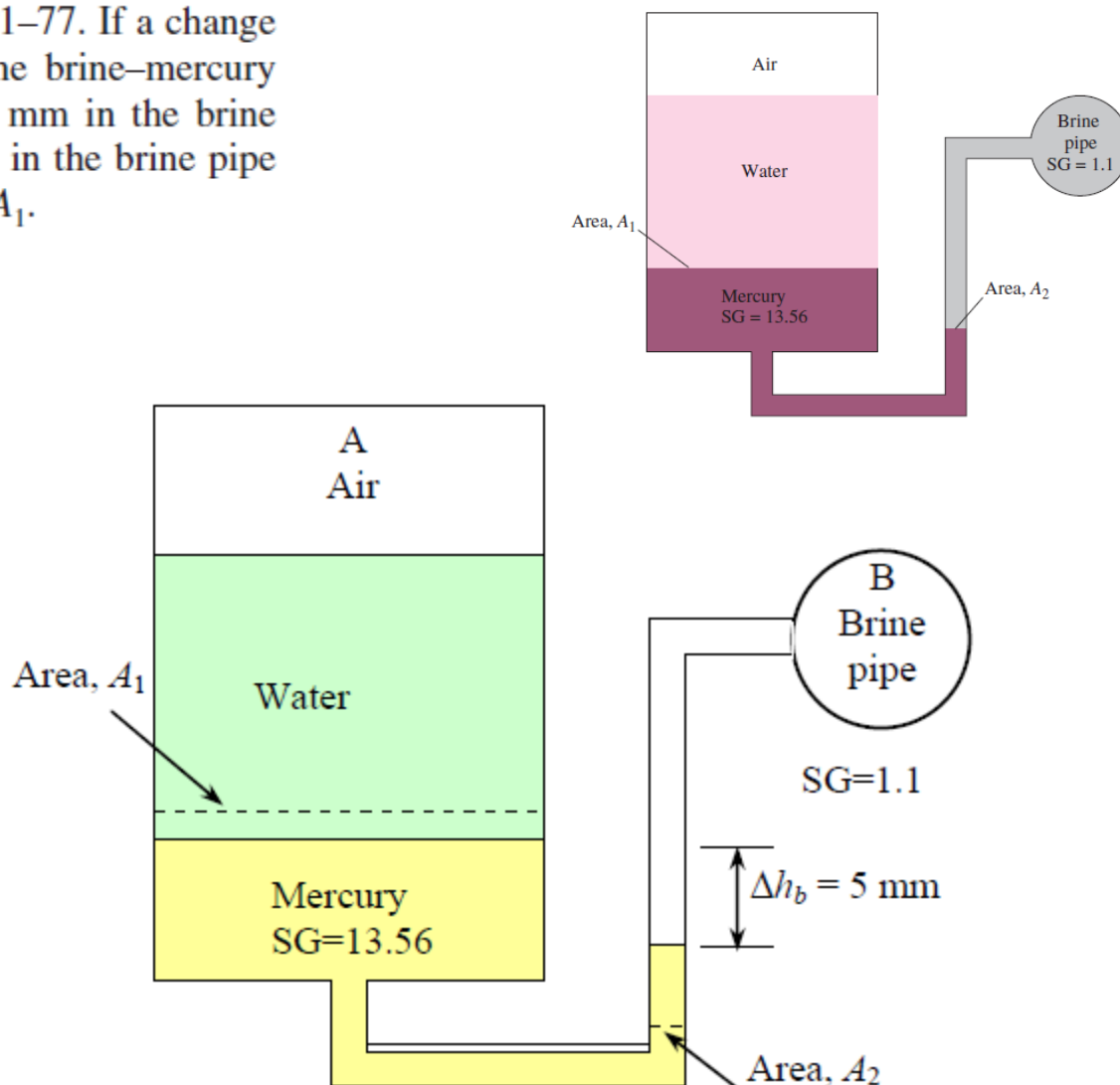
1-77 Consider the system shown in Fig. P1-77. If a change of 0.7 kPa in the pressure of air causes the brine–mercury interface in the right column to drop by 5 mm in the brine level in the right column while the pressure in the brine pipe remains constant, determine the ratio of A_2/A_1 .

Assumptions 1 All the liquids are incompressible. 2 Pressure in the brine pipe remains constant. 3 The variation of pressure in the trapped air space is negligible.

Properties The specific gravities are given to be 13.56 for mercury and 1.1 for brine. We take the standard density of water to be $\rho_w = 1000 \text{ kg/m}^3$.

Analysis It is clear from the problem statement and the figure that the brine pressure is much higher than the air pressure, and when the air pressure drops by 0.7 kPa, the pressure difference between the brine and the air space increases also by the same amount.

Starting with the air pressure (point A) and moving along the tube by adding (as we go down) or subtracting (as we go up) the ρgh terms until we reach the brine pipe (point B), and setting the result equal to P_B before and after the pressure change of air give



1-77 Continued;

$$\text{Before: } P_{A1} + \rho_w g h_w + \rho_{\text{Hg}} g h_{\text{Hg},1} - \rho_{\text{br}} g h_{\text{br},1} = P_B$$

$$\text{After: } P_{A2} + \rho_w g h_w + \rho_{\text{Hg}} g h_{\text{Hg},2} - \rho_{\text{br}} g h_{\text{br},2} = P_B$$

Subtracting,

$$P_{A2} - P_{A1} + \rho_{\text{Hg}} g \Delta h_{\text{Hg}} - \rho_{\text{br}} g \Delta h_{\text{br}} = 0 \rightarrow \frac{P_{A1} - P_{A2}}{\rho_w g} = S G_{\text{Hg}} \Delta h_{\text{Hg}} - S G_{\text{br}} \Delta h_{\text{br}} = 0 \quad (1)$$

where Δh_{Hg} and Δh_{br} are the changes in the differential mercury and brine column heights, respectively, due to the drop in air pressure. Both of these are positive quantities since as the mercury-brine interface drops, the differential fluid heights for both mercury and brine increase. Noting also that the volume of mercury is constant, we have $A_1 \Delta h_{\text{Hg, left}} = A_2 \Delta h_{\text{Hg, right}}$ and

$$P_{A2} - P_{A1} = -0.7 \text{ kPa} = -700 \text{ N/m}^2 = -700 \text{ kg/m} \cdot \text{s}^2$$

$$\Delta h_{\text{br}} = 0.005 \text{ m}$$

$$\Delta h_{\text{Hg}} = \Delta h_{\text{Hg, right}} + \Delta h_{\text{Hg, left}} = \Delta h_{\text{br}} + \Delta h_{\text{br}} A_2 / A_1 = \Delta h_{\text{br}} (1 + A_2 / A_1)$$

Substituting,

$$\frac{700 \text{ kg/m} \cdot \text{s}^2}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} = [13.56 \times 0.005(1 + A_2 / A_1) - (1.1 \times 0.005)] \text{ m}$$

It gives

$$A_2 / A_1 = \mathbf{0.134}$$

1-78 A multifluid container is connected to a U-tube, as shown in Fig. P1-78. For the given specific gravities and fluid column heights, determine the gage pressure at A. Also determine the height of a mercury column that would create the same pressure at A. **Answers:** 0.471 kPa, 0.353 cm

Analysis Starting with the atmospheric pressure on the top surface of the container and moving along the tube by adding (as we go down) or subtracting (as we go up) the ρgh terms until we reach point A, and setting the result equal to P_A give

$$P_{\text{atm}} + \rho_{\text{oil}}gh_{\text{oil}} + \rho_w gh_w - \rho_{\text{gly}}gh_{\text{gly}} = P_A$$

Rearranging and using the definition of specific gravity,

$$P_A - P_{\text{atm}} = SG_{\text{oil}}\rho_w gh_{\text{oil}} + SG_w \rho_w gh_w - SG_{\text{gly}}\rho_w gh_{\text{gly}}$$

or

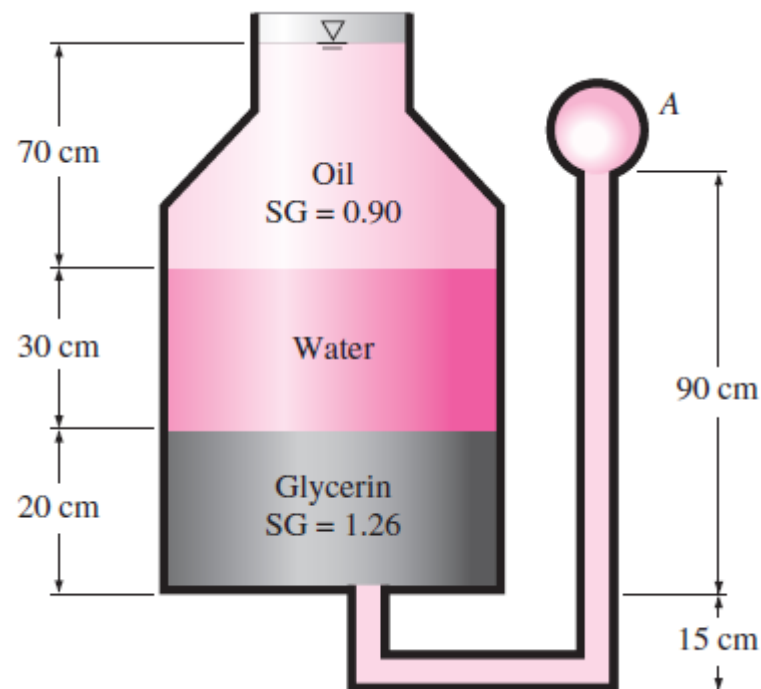
$$P_{A,\text{gage}} = g\rho_w (SG_{\text{oil}}h_{\text{oil}} + SG_w h_w - SG_{\text{gly}}h_{\text{gly}})$$

Substituting,

$$\begin{aligned} P_{A,\text{gage}} &= (9.81 \text{ m/s}^2)(1000 \text{ kg/m}^3)[0.90(0.70 \text{ m}) + 1(0.3 \text{ m}) - 1.26(0.70 \text{ m})] \left(\frac{1 \text{ kN}}{1000 \text{ kg} \cdot \text{m/s}^2} \right) \\ &= 0.471 \text{ kN/m}^2 = \mathbf{0.471 \text{ kPa}} \end{aligned}$$

The equivalent mercury column height is

$$h_{\text{Hg}} = \frac{P_{A,\text{gage}}}{\rho_{\text{Hg}}g} = \frac{0.471 \text{ kN/m}^2}{(13,600 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \left(\frac{1000 \text{ kg} \cdot \text{m/s}^2}{1 \text{ kN}} \right) = 0.00353 \text{ m} = \mathbf{0.353 \text{ cm}}$$



Discussion Note that the high density of mercury makes it a very suitable fluid for measuring high pressures in manometers.

Chapter 1

End of the Problems